

Comment & Response

Artificial Intelligence–Based Detection of Diabetic Retinopathy

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To the Editor In the article titled “Use of Artificial Intelligence–Based Detection of Diabetic Retinopathy in the US,”¹ Shah et al noted the following observations based on an analysis using the TriNetX database: “Since 2021, AI [artificial intelligence] imaging (92229) was used in 3440 of 154136 cases (2.2%), meaning that only 0.09% of all patients with diabetes received this new modality for DR [diabetic retinopathy] detection,” and “this potential underuse of AI for DR screening may be further pronounced....”

One explanation for the observation of potential underuse is that AI-based detection services may have been performed but were not documented using the assigned *Current Procedural Terminology (CPT)* code in this period. The reimbursement rates are relatively low and there may be obstacles for payment and no existing billing framework, so services might not be billed by the organizations or clinicians performing the screening.²

Another explanation might be that AI-based detection services are not documented across the entire patient journey. The source database was the TriNetX database, which consisted of 62 large institutions across the United States, but it has not been established that these patients receive AI-based detection services or even all their eye care services at these large institutions. Table 1 in the article by Shah and colleagues indicates that the rate for all ocular imaging services (*CPT* codes 92227, 92228, 92229, 92250 and 92134) across all patients with diabetes ranged from 2.8% to 3.5% from 2019 to 2023, which appears to be low compared with the expected rate. A previously published study³ on the prevalence of diabetes-related eye disease estimated that 26.43% (95% uncertainty interval [UI], 21.95%-31.60%) of all patients in the United States with diabetes had diabetic retinopathy, and 5.06% (95% UI, 3.90-6.57) had vision-threatening diabetic retinopathy in 2021. The US Food and Drug Administration (FDA) describes the following for data source selection for real-world data considerations: “A crucial step in selecting a data source is determining whether it

gree a data source captures the outcome of interest should be assessed before study initiation and be independent of the exposure of interest."⁴ This study limitation does not appear to be described in the article by Shah and colleagues.

There are 3 AI-based detection devices for diabetic retinopathy, rather than only 2 that are mentioned by Shah and colleagues, with the third device, the AEYE-DS (AEYE Health) being approved by the FDA in November 2022.⁵

Article Information

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